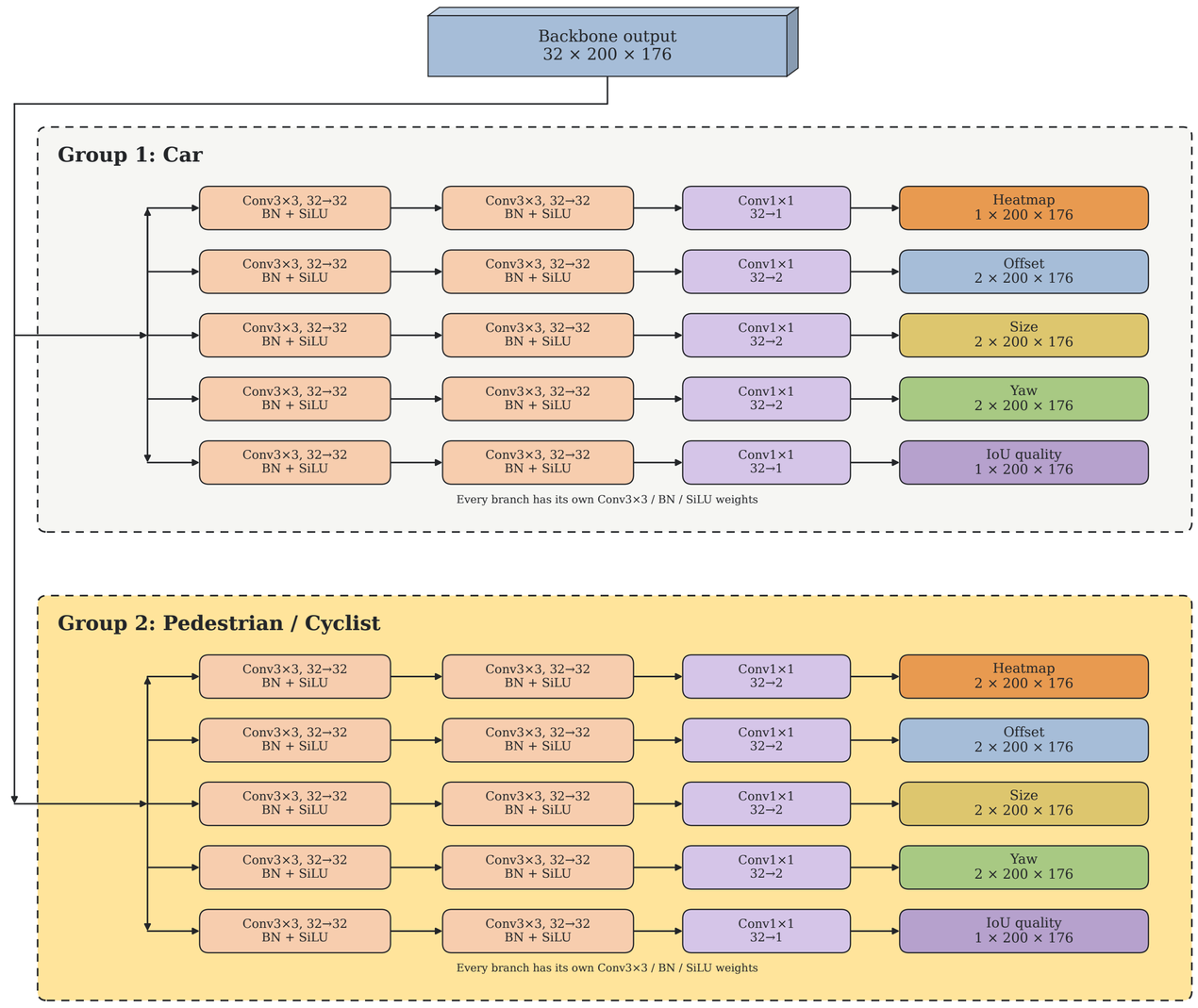
Grouped detection heads · current vs proposed

A · Current code

Two private Conv3×3 layers in every prediction branch



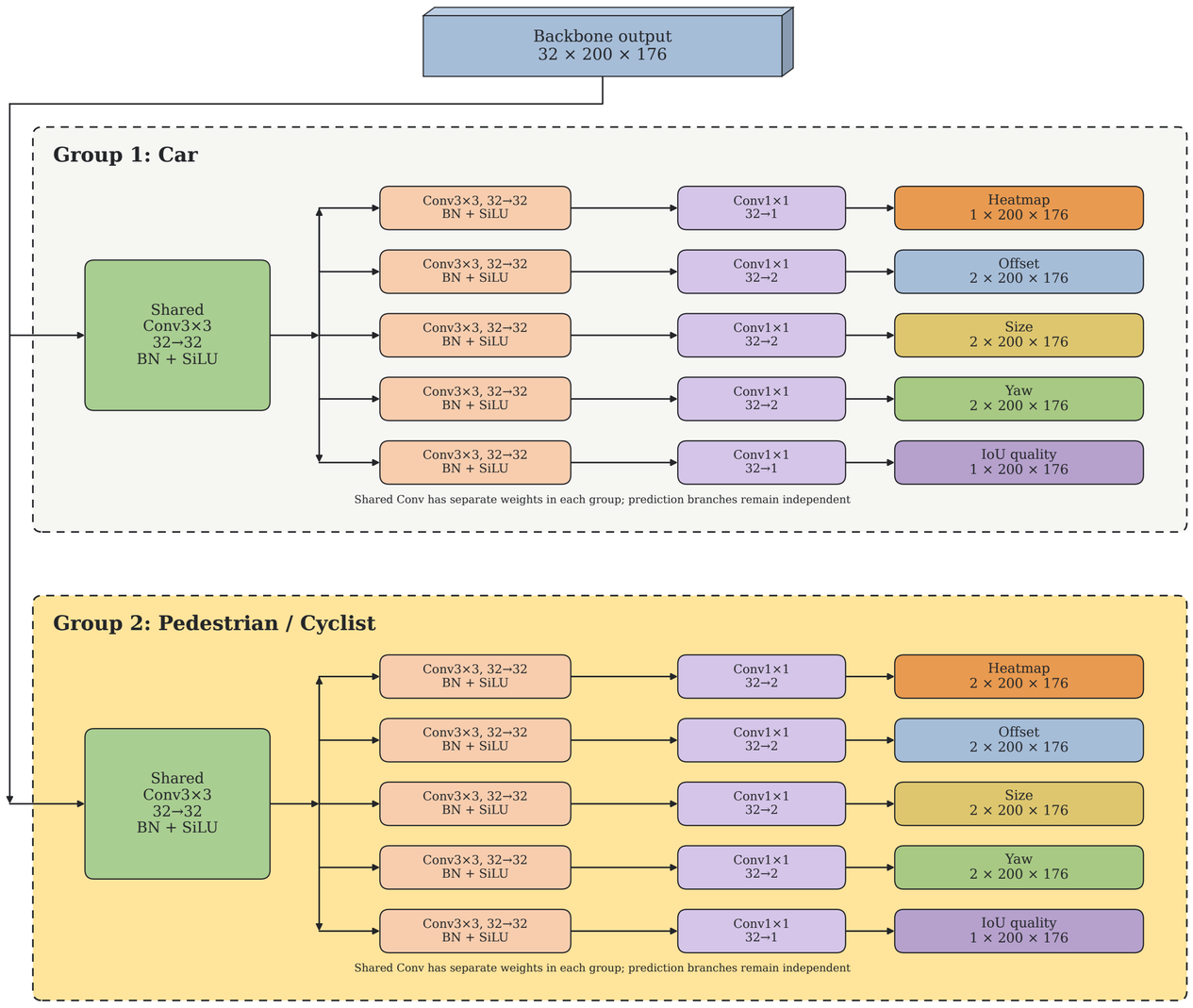
20 Conv3×3 · 10 Conv1×1 · 186,161 head parameters

Convolution cost: 6.51 G MAC / sample

Topology taken directly from GroupedHeader and Header / Head

B · Proposed head

One shared Conv3×3 per group + one private Conv3×3 per branch



12 Conv3×3 · 10 Conv1×1 · 111,921 head parameters

Convolution cost: 3.91 G MAC / sample

Implemented: header_architecture=shared; training results pending

Shapes: C × H × W. Both groups receive all 32 channels. MAC counts include convolutions only; backbone is excluded.